

설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
19.5	27.5	28.8	0.3	0.15	11	25	56	119	15	21	30	43	3	0	− 5	− 12	0.017
19.5	27.5	28.8	0.3	0.15	10	25	59	131	16	23	33	49	3	0	− 5	− 12	0.015
19.5	27.5	28.8	0.3	0.15	15	35	77	149	34	47	63	82	1	− 1	− 4	− 8	0.017
19.5	27.5	28.8	0.3	0.15	14	37	84	167	37	53	72	95	1	− 1	− 4	− 8	0.015
19.5	27.5	28.8	0.3	0.15	12	25	46	–	14	19	24	–	3	0	− 4	–	0.018
19.5	27.5	28.8	0.3	0.15	11	25	49	–	13	21	27	–	3	0	− 4	–	0.016
19.5	27.5	28.8	0.3	0.15	11	25	49	–	13	21	27	–	3	0	− 4	–	0.016
19.5	32.5	33.8	0.3	0.15	15	30	69	156	16	21	31	46	2	− 1	− 7	− 16	0.039
19.5	32.5	33.8	0.3	0.15	15	30	75	174	18	24	35	53	2	− 1	− 7	− 16	0.036
19.5	32.5	33.8	0.3	0.15	25	47	127	257	41	51	74	99	0	− 2	− 7	− 13	0.040
19.5	32.5	33.8	0.3	0.15	25	50	142	293	46	58	86	116	0	− 2	− 7	− 13	0.037
19.5	32.5	33.8	0.3	0.15	25	116	251	422	54	92	123	151	0	− 5	− 10	− 15	0.040
19.5	32.5	33.8	0.3	0.15	13	29	60	–	13	18	24	–	3	− 1	− 7	–	0.039
19.5	32.5	33.8	0.3	0.15	12	29	65	–	14	20	28	–	3	− 1	− 7	–	0.036
19.5	32.5	33.8	0.3	0.15	12	29	65	–	14	20	28	–	3	− 1	− 7	–	0.036
22	35	37.5	0.6	0.3	25	46	146	296	19	25	41	59	0	− 4	− 16	− 28	0.065
22	35	37.5	0.6	0.3	25	49	163	337	21	28	48	69	0	− 4	− 16	− 28	0.058
22	35	37.5	0.6	0.3	35	75	204	408	45	60	87	116	− 1	− 4	− 11	− 19	0.064
22	35	37.5	0.6	0.3	37	82	232	470	51	69	102	136	− 1	− 4	− 11	− 19	0.057
22	35	37.5	0.6	0.3	25	115	247	412	53	90	119	145	0	− 5	− 10	− 15	0.065
22	35	37.5	0.6	0.3	25	48	97	–	16	21	28	–	0	− 5	− 13	–	0.065
22	35	37.5	0.6	0.3	25	51	107	–	18	24	32	–	0	− 5	− 13	–	0.057
22	35	37.5	0.6	0.3	25	51	107	–	18	24	32	–	0	− 5	− 13	–	0.057
22.5	34.5	35.8	0.3	0.15	20	42	80	152	19	26	35	48	1	− 3	− 8	− 15	0.036
22.5	34.5	35.8	0.3	0.15	19	44	87	169	21	29	40	55	1	− 3	− 8	− 15	0.033
22.5	34.5	35.8	0.3	0.15	25	63	114	247	43	60	75	102	0	− 3	− 6	− 12	0.037
22.5	34.5	35.8	0.3	0.15	25	68	127	282	48	69	87	119	0	− 3	− 6	− 12	0.034
22.5	34.5	35.8	0.3	0.15	16	29	63	–	15	19	26	–	2	− 1	− 7	–	0.036
22.5	34.5	35.8	0.3	0.15	15	30	67	–	17	21	29	–	2	− 1	− 7	–	0.033
22.5	34.5	35.8	0.3	0.15	15	30	67	–	17	21	29	–	2	− 1	− 7	–	0.033
25	37	39.5	0.6	0.3	25	49	119	244	21	28	42	59	0	− 4	− 12	− 22	0.067
25	37	39.5	0.6	0.3	25	52	132	277	24	32	48	68	0	− 4	− 12	− 22	0.060
25	37	39.5	0.6	0.3	36	81	206	403	51	68	97	127	− 1	− 4	− 10	− 17	0.067
25	37	39.5	0.6	0.3	38	90	234	465	58	79	113	149	− 1	− 4	− 10	− 17	0.060
25	37	39.5	0.6	0.3	25	128	280	473	59	104	139	170	0	− 5	− 10	− 15	0.068
25	37	39.5	0.6	0.3	25	51	107	–	18	24	32	–	0	− 5	− 13	–	0.068
25	37	39.5	0.6	0.3	25	54	119	–	20	27	37	–	0	− 5	− 13	–	0.061
25	37	39.5	0.6	0.3	25	54	119	–	20	27	37	–	0	− 5	− 13	–	0.061
26	41	42	1	0.5	35	68	196	384	23	30	48	68	− 2	− 7	− 20	− 33	0.103
26	41	42	1	0.5	37	74	221	440	26	34	56	79	− 2	− 7	− 20	− 33	0.091
26	41	42	1	0.5	63	112	290	596	59	73	104	140	− 3	− 6	− 14	− 24	0.102
26	41	42	1	0.5	68	125	332	691	67	84	122	165	− 3	− 6	− 14	− 24	0.090
26	41	42	1	0.5	25	262	439	649	56	128	155	181	0	− 10	− 15	− 20	0.104
26	41	42	1	0.5	34	67	140	–	19	25	34	–	− 2	− 8	− 18	–	0.103
26	41	42	1	0.5	35	72	157	–	22	29	39	–	− 2	− 8	− 18	–	0.091
26	41	42	1	0.5	35	72	157	–	22	29	39	–	− 2	− 8	− 18	–	0.091

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
25°	2.0			
30°	1.4			

B표

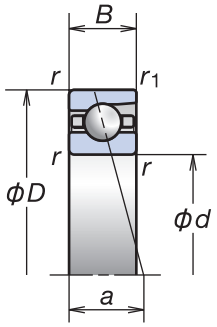
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

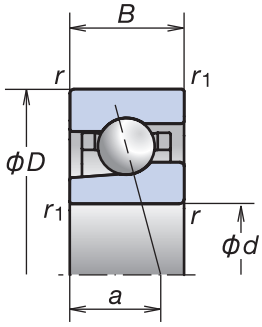
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

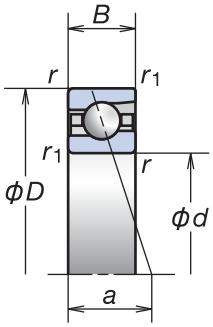
내경25mm



79,70,72계열



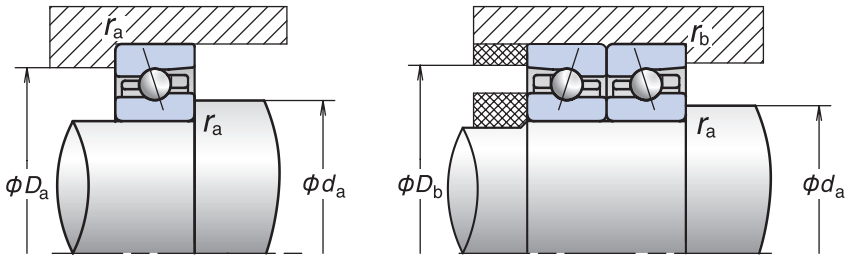
BSR19,10,02계열



BNR19계열
BER19계열

호칭 번호[1]	주요 치수 (mm)					기본 정격 하중[2] (kN)		허용[3] 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수[4] (min^{-1})	
	d	D	B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
7905C	**25**	**42**	**9**	0.3	0.15	8.25	5.40	3.90	15	15.5	9.0	34 400	52 300
7905CSN24	**25**	**42**	**9**	0.3	0.15	(8.25)	(5.40)	4.63	15	15.5	9.0	44 800	68 100
7905A5	**25**	**42**	**9**	0.3	0.15	7.80	5.15	4.40	25	–	12.3	29 900	44 800
7905A5SN24	**25**	**42**	**9**	0.3	0.15	(7.80)	(5.15)	5.20	25	–	12.3	38 900	58 300
***25BSR19S**	**25**	**42**	**9**	0.3	0.15	5.30	2.71	3.40	15	7.8	9.0	47 800	65 700
***25BSR19H**	**25**	**42**	**9**	0.3	0.15	(5.30)	(2.71)	2.22	15	7.8	9.0	56 800	83 600
***25BSR19X**	**25**	**42**	**9**	0.3	0.15	(5.30)	(2.71)	2.22	15	7.8	9.0	65 700	89 600
7005C	**25**	**47**	**12**	0.6	0.3	12.3	7.40	5.20	15	14.7	10.8	32 000	48 700
7005CSN24	**25**	**47**	**12**	0.6	0.3	(12.3)	(7.40)	6.16	15	14.7	10.8	41 700	63 400
7005A5	**25**	**47**	**12**	0.6	0.3	11.7	7.10	5.95	25	–	14.4	27 800	41 700
7005A5SN24	**25**	**47**	**12**	0.6	0.3	(11.7)	(7.10)	7.08	25	–	14.4	36 200	54 200
7005A	**25**	**47**	**12**	0.6	0.3	11.3	6.85	4.55	30	–	16.4	20 900	27 800
***25BSR10S**	**25**	**47**	**12**	0.6	0.3	7.90	3.75	4.65	15	7.6	10.8	44 500	61 200
***25BSR10H**	**25**	**47**	**12**	0.6	0.3	(7.90)	(3.75)	3.05	15	7.6	10.8	52 800	77 800
***25BSR10X**	**25**	**47**	**12**	0.6	0.3	(7.90)	(3.75)	3.05	15	7.6	10.8	61 200	83 400
7205C	**25**	**52**	**15**	1	0.6	17.4	10.2	7.50	15	14.0	12.7	29 900	45 500
7205CSN24	**25**	**52**	**15**	1	0.6	(17.4)	(10.2)	8.91	15	14.0	12.7	39 000	59 300
7205A5	**25**	**52**	**15**	1	0.6	16.7	9.80	9.05	25	–	16.5	26 000	39 000
7205A5SN24	**25**	**52**	**15**	1	0.6	(16.7)	(9.80)	10.7	25	–	16.5	33 800	50 700
7205A	**25**	**52**	**15**	1	0.6	16.1	9.45	6.95	30	–	18.6	19 500	26 000
***25BSR02S**	**25**	**52**	**15**	1	0.6	11.0	5.20	6.45	15	7.1	12.7	41 600	57 200
***25BSR02H**	**25**	**52**	**15**	1	0.6	(11.0)	(5.20)	4.20	15	7.1	12.7	49 400	72 800
***25BSR02X**	**25**	**52**	**15**	1	0.6	(11.0)	(5.20)	4.20	15	7.1	12.7	57 200	78 000

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다.
(2) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(3) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(4) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
27.5	39.5	40.8	0.3	0.15	19	37	99	203	21	28	43	61	1	− 2	− 9	− 17	0.043
27.5	39.5	40.8	0.3	0.15	18	39	109	229	23	31	49	70	1	− 2	− 9	− 17	0.039
27.5	39.5	40.8	0.3	0.15	38	70	153	290	57	71	96	124	− 1	− 3	− 7	− 12	0.043
27.5	39.5	40.8	0.3	0.15	39	76	172	332	64	81	111	144	− 1	− 3	− 7	− 12	0.039
27.5	39.5	40.8	0.3	0.15	20	41	76	–	18	25	32	–	1	− 3	− 8	–	0.043
27.5	39.5	40.8	0.3	0.15	19	43	83	–	20	28	36	–	1	− 3	− 8	–	0.039
27.5	39.5	40.8	0.3	0.15	19	43	83	–	20	28	36	–	1	− 3	− 8	–	0.039
30	42	44.5	0.6	0.3	30	58	148	292	24	32	48	67	− 1	− 5	− 14	− 24	0.078
30	42	44.5	0.6	0.3	31	62	165	332	27	36	55	78	− 1	− 5	− 14	− 24	0.070
30	42	44.5	0.6	0.3	52	104	193	397	61	79	100	133	− 2	− 5	− 9	− 16	0.077
30	42	44.5	0.6	0.3	55	116	220	458	70	91	116	156	− 2	− 5	− 9	− 16	0.069
30	42	44.5	0.6	0.3	25	135	299	507	63	112	149	183	0	− 5	− 10	− 15	0.079
30	42	44.5	0.6	0.3	25	52	112	–	20	25	35	–	0	− 5	− 13	–	0.078
30	42	44.5	0.6	0.3	25	56	125	–	22	29	40	–	0	− 5	− 13	–	0.070
30	42	44.5	0.6	0.3	25	56	125	–	22	29	40	–	0	− 5	− 13	–	0.070
31	46	47	1	0.5	42	82	193	402	27	36	53	76	1	− 4	− 14	− 27	0.127
31	46	47	1	0.5	41	86	212	452	30	41	61	88	1	− 4	− 14	− 27	0.112
31	46	47	1	0.5	82	143	330	691	73	89	123	166	− 2	− 5	− 12	− 22	0.130
31	46	47	1	0.5	87	156	372	793	83	103	143	194	− 2	− 5	− 12	− 22	0.115
31	46	47	1	0.5	49	357	578	839	80	161	193	223	0	− 10	− 15	− 20	0.129
31	46	47	1	0.5	37	84	163	–	22	31	40	–	2	− 5	− 14	–	0.127
31	46	47	1	0.5	36	88	179	–	25	35	46	–	2	− 5	− 14	–	0.112
31	46	47	1	0.5	36	88	179	–	25	35	46	–	2	− 5	− 14	–	0.112

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

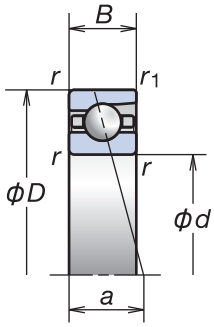
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

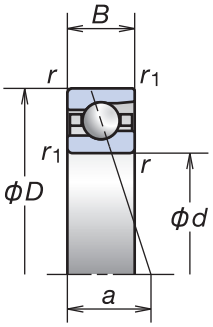
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

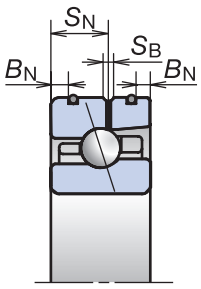
내경30mm



79,70,72계열
BNR10,20계열
BER10,20계열



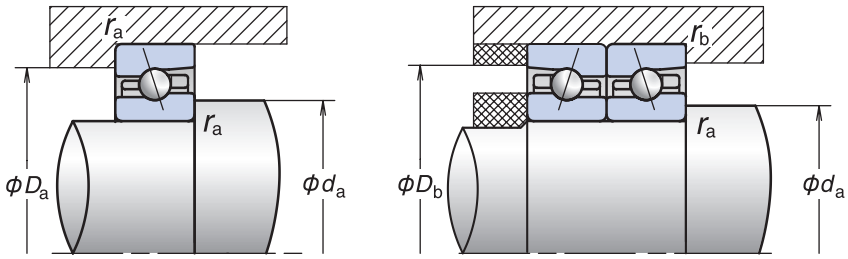
BNR19계열
BER19계열



로바스트 쇼트

호칭 번호(1)	주요 치수(2) (mm)								기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7906C	30	47	9	–	–	–	0.3	0.15	8.70	6.25	4.40	15	15.9	9.7	29 900	45 500
*7906CSN24	30	47	9	–	–	–	0.3	0.15	(8.70)	(6.25)	5.20	15	15.9	9.7	39 000	59 300
*7906A5	30	47	9	–	–	–	0.3	0.15	8.25	5.95	4.95	25	–	13.5	26 000	39 000
*7906A5SN24	30	47	9	–	–	–	0.3	0.15	(8.25)	(5.95)	5.86	25	–	13.5	33 800	50 700
*30BN19BV1V	30	47	9	–	–	–	0.3	0.15	5.00	3.60	5.05	18	10.9	10.8	36 400	–
*30BN19BSN24V1V	30	47	9	–	–	–	0.3	0.15	(5.00)	(3.60)	3.30	18	10.9	10.8	46 800	–
*30BA19BV1V	30	47	9	–	–	–	0.3	0.15	4.80	3.45	5.90	25	–	13.5	31 200	–
*30BA19BSN24V1V	30	47	9	–	–	–	0.3	0.15	(4.80)	(3.45)	4.00	25	–	13.5	41 600	–
30BNR19S	30	47	9	–	–	–	0.3	0.15	6.30	4.05	5.75	18	10.5	10.8	36 400	52 000
30BNR19H	30	47	9	1.1	5.6	1.4	0.3	0.15	(6.30)	(4.05)	3.80	18	10.5	10.8	46 800	72 800
30BNR19X	30	47	9	1.1	5.6	1.4	0.3	0.15	(6.30)	(4.05)	3.80	18	10.5	10.8	54 600	85 800
30BER19S	30	47	9	–	–	–	0.3	0.15	6.00	3.90	6.80	25	–	13.5	31 200	44 200
30BER19H	30	47	9	1.1	5.6	1.4	0.3	0.15	(6.00)	(3.90)	4.60	25	–	13.5	41 600	65 000
30BER19X	30	47	9	1.1	5.6	1.4	0.3	0.15	(6.00)	(3.90)	4.60	25	–	13.5	49 400	78 000
*7006C	30	55	13	–	–	–	1	0.6	15.9	10.3	6.85	15	14.9	12.2	27 100	41 200
*7006CSN24	30	55	13	–	–	–	1	0.6	(15.9)	(10.3)	8.12	15	14.9	12.2	35 300	53 700
*7006A5	30	55	13	–	–	–	1	0.6	15.1	9.80	8.05	25	–	16.4	23 600	35 300
*7006A5SN24	30	55	13	–	–	–	1	0.6	(15.1)	(9.80)	9.56	25	–	16.4	30 600	45 900
*7006A	30	55	13	–	–	–	1	0.6	14.6	9.45	6.20	30	–	18.8	17 700	23 600
*30BNR10S	30	55	13	–	–	–	1	0.6	8.65	5.75	8.20	18	10.3	13.3	33 000	47 100
*30BNR10H	30	55	13	2.8	7.5	1.4	1	0.6	(8.65)	(5.75)	5.35	18	10.3	13.3	42 400	65 900
*30BNR10X	30	55	13	2.8	7.5	1.4	1	0.6	(8.65)	(5.75)	5.35	18	10.3	13.3	49 500	77 700
*30BER10S	30	55	13	–	–	–	1	0.6	8.30	5.50	9.65	25	–	16.3	28 300	40 000
*30BER10H	30	55	13	2.8	7.5	1.4	1	0.6	(8.30)	(5.50)	6.50	25	–	16.3	37 700	58 900
*30BER10X	30	55	13	2.8	7.5	1.4	1	0.6	(8.30)	(5.50)	6.50	25	–	16.3	44 800	70 600
*30BNR20SV1V	30	55	16	–	–	–	1	0.6	8.65	5.75	8.20	18	10.3	14.8	33 000	–
*30BNR20HV1V	30	55	16	–	–	–	1	0.6	(8.65)	(5.75)	5.35	18	10.3	14.8	42 400	–
*30BNR20XV1V	30	55	16	–	–	–	1	0.6	(8.65)	(5.75)	5.35	18	10.3	14.8	49 500	–
*30BER20SV1V	30	55	16	–	–	–	1	0.6	8.30	5.50	9.65	25	–	17.8	28 300	–
*30BER20HV1V	30	55	16	–	–	–	1	0.6	(8.30)	(5.50)	6.50	25	–	17.8	37 700	–
*30BER20XV1V	30	55	16	–	–	–	1	0.6	(8.30)	(5.50)	6.50	25	–	17.8	44 800	–
7206C	30	62	16	–	–	–	1	0.6	24.2	14.7	10.3	15	13.9	14.2	25 000	38 100
7206CSN24	30	62	16	–	–	–	1	0.6	(24.2)	(14.7)	12.2	15	13.9	14.2	32 700	49 600
7206A5	30	62	16	–	–	–	1	0.6	23.2	14.1	12.0	25	–	18.7	21 800	32 700
7206A5SN24	30	62	16	–	–	–	1	0.6	(23.2)	(14.1)	14.2	25	–	18.7	28 300	42 400
7206A	30	62	16	–	–	–	1	0.6	22.4	13.6	9.20	30	–	21.3	16 400	21 800

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(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
32.5	44.5	45.8	0.3	0.15	25	46	95	204	25	33	45	65	0	− 3	− 8	− 16	0.049
32.5	44.5	45.8	0.3	0.15	25	49	104	230	28	37	51	75	0	− 3	− 8	− 16	0.044
32.5	44.5	45.8	0.3	0.15	39	74	141	285	62	78	99	131	− 1	− 3	− 6	− 11	0.050
32.5	44.5	45.8	0.3	0.15	41	81	158	326	70	90	115	154	− 1	− 3	− 6	− 11	0.045
32.5	44.5	45.8	0.3	0.15	25	109	218	–	30	53	71	–	0	− 8	− 15	–	0.050
32.5	44.5	45.8	0.3	0.15	25	121	248	–	34	62	83	–	0	− 8	− 15	–	0.047
32.5	44.5	45.8	0.3	0.15	25	178	352	–	50	100	131	–	0	− 8	− 14	–	0.050
32.5	44.5	45.8	0.3	0.15	25	202	405	–	56	117	153	–	0	− 8	− 14	–	0.047
32.5	44.5	45.8	0.3	0.15	25	101	197	–	28	48	63	–	0	− 8	− 15	–	0.048
32.5	44.5	45.8	0.3	0.15	25	112	224	–	31	55	73	–	0	− 8	− 15	–	0.043
32.5	44.5	45.8	0.3	0.15	25	112	224	–	31	55	73	–	0	− 8	− 15	–	0.043
32.5	44.5	45.8	0.3	0.15	25	164	318	–	47	90	116	–	0	− 8	− 14	–	0.048
32.5	44.5	45.8	0.3	0.15	25	185	366	–	52	105	136	–	0	− 8	− 14	–	0.043
32.5	44.5	45.8	0.3	0.15	25	185	366	–	52	105	136	–	0	− 8	− 14	–	0.043
36	49	50	1	0.5	41	75	195	386	30	38	58	81	1	− 3	− 13	− 24	0.114
36	49	50	1	0.5	41	78	214	434	33	42	66	94	1	− 3	− 13	− 24	0.102
36	49	50	1	0.5	66	129	294	590	73	93	127	169	− 1	− 4	− 10	− 18	0.114
36	49	50	1	0.5	68	139	331	676	82	106	148	197	− 1	− 4	− 10	− 18	0.102
36	49	50	1	0.5	49	191	390	638	87	138	180	217	0	− 5	− 10	− 15	0.116
36	49	50	1	0.5	49	106	229	–	39	52	71	–	0	− 5	− 13	–	0.124
36	49	50	1	0.5	49	113	254	–	44	59	82	–	0	− 5	− 13	–	0.116
36	49	50	1	0.5	49	113	254	–	44	59	82	–	0	− 5	− 13	–	0.116
36	49	50	1	0.5	49	224	443	–	64	109	141	–	0	− 8	− 15	–	0.124
36	49	50	1	0.5	49	250	505	–	72	126	165	–	0	− 8	− 15	–	0.116
36	49	50	1	0.5	49	250	505	–	72	126	165	–	0	− 8	− 15	–	0.116
36	49	50	1	0.5	49	106	229	–	39	52	71	–	0	− 5	− 13	–	0.150
36	49	50	1	0.5	49	113	254	–	44	59	82	–	0	− 5	− 13	–	0.142
36	49	50	1	0.5	49	113	254	–	44	59	82	–	0	− 5	− 13	–	0.142
36	49	50	1	0.5	49	224	443	–	64	109	141	–	0	− 8	− 15	–	0.150
36	49	50	1	0.5	49	250	505	–	72	126	165	–	0	− 8	− 15	–	0.142
36	49	50	1	0.5	49	250	505	–	72	126	165	–	0	− 8	− 15	–	0.142
36	56	57	1	0.5	57	114	292	591	33	43	66	94	− 1	− 7	− 20	− 35	0.194
36	56	57	1	0.5	58	122	326	673	36	49	76	109	− 1	− 7	− 20	− 35	0.169
36	56	57	1	0.5	105	202	457	881	85	108	147	192	− 3	− 7	− 15	− 25	0.194
36	56	57	1	0.5	113	224	521	1 018	97	124	172	225	− 3	− 7	− 15	− 25	0.169
36	56	57	1	0.5	49	384	625	908	86	175	210	243	0	− 10	− 15	− 20	0.197

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

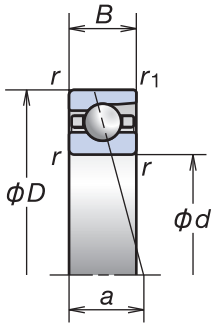
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

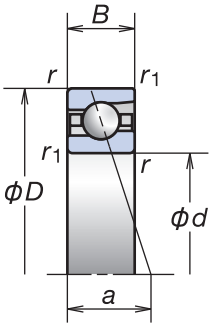
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

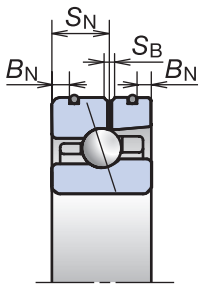
내경35mm



79,70,72계열
BNR10,20계열
BER10,20계열



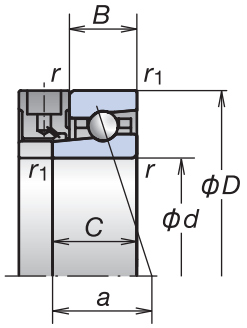
BNR19계열
BER19계열



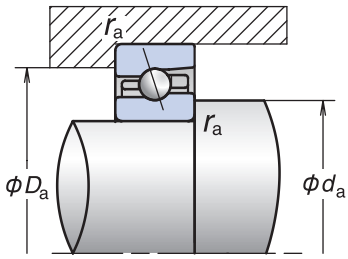
로바스트 쇼트

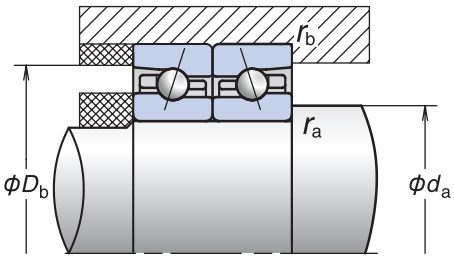
호칭 번호(1)	주요 치수(2) (mm)								기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7907C	35	55	10	–	–	–	0.6	0.3	12.7	9.15	6.60	15	15.7	11.0	25 600	38 900
*7907CSN24	35	55	10	–	–	–	0.6	0.3	(12.7)	(9.15)	7.82	15	15.7	11.0	33 400	50 700
*7907A5	35	55	10	–	–	–	0.6	0.3	12.0	8.70	7.20	25	-	15.5	22 300	33 400
*7907A5SN24	35	55	10	–	–	–	0.6	0.3	(12.0)	(8.70)	8.52	25	-	15.5	28 900	43 400
*35BN19AV1V	35	55	10	–	–	–	0.6	0.3	6.95	5.10	7.20	18	10.8	12.3	31 200	–
*35BN19ASN24V1V	35	55	10	–	–	–	0.6	0.3	(6.95)	(5.10)	4.75	18	10.8	12.3	40 000	–
*35BA19BV1V	35	55	10	–	–	–	0.6	0.3	6.65	4.90	8.50	25	–	15.5	26 700	–
*35BA19BSN24V1V	35	55	10	–	–	–	0.6	0.3	(6.65)	(4.90)	5.75	25	–	15.5	35 600	–
35BNR19S	35	55	10	–	–	–	0.6	0.3	9.20	6.00	8.55	18	10.4	12.3	31 200	44 500
35BNR19H	35	55	10	1.6	6.1	1.4	0.6	0.3	(9.20)	(6.00)	5.60	18	10.4	12.3	40 000	62 300
35BNR19X	35	55	10	1.6	6.1	1.4	0.6	0.3	(9.20)	(6.00)	5.60	18	10.4	12.3	46 700	73 400
35BER19S	35	55	10	–	–	–	0.6	0.3	8.80	5.75	10.0	25	-	15.5	26 700	37 800
35BER19H	35	55	10	1.6	6.1	1.4	0.6	0.3	(8.80)	(5.75)	6.80	25	-	15.5	35 600	55 600
35BER19X	35	55	10	1.6	6.1	1.4	0.6	0.3	(8.80)	(5.75)	6.80	25	-	15.5	42 300	66 700
*7007C	35	62	14	–	–	–	1	0.6	20.1	13.7	9.35	15	15.0	13.5	23 800	36 100
*7007CSN24	35	62	14	–	–	–	1	0.6	(20.1)	(13.7)	11.1	15	15.0	13.5	29 700	45 200
*7007A5	35	62	14	–	–	–	1	0.6	19.1	13.0	11.4	25	-	18.3	20 700	31 000
*7007A5SN24	35	62	14	–	–	–	1	0.6	(19.1)	(13.0)	13.5	25	-	18.3	25 800	38 800
*7007A	35	62	14	–	–	–	1	0.6	18.4	12.6	8.75	30	-	21.0	15 500	20 700
*35BNR10S	35	62	14	–	–	–	1	0.6	10.1	7.10	10.2	18	10.6	14.8	28 900	41 300
*35BNR10H	35	62	14	2.8	8.3	1.4	1	0.6	(10.1)	(7.10)	6.70	18	10.6	14.8	37 200	57 800
*35BNR10X	35	62	14	2.8	8.3	1.4	1	0.6	(10.1)	(7.10)	6.70	18	10.6	14.8	43 300	68 100
*35BER10S	35	62	14	–	–	–	1	0.6	9.70	6.85	12.0	25	-	18.2	24 800	35 100
*35BER10H	35	62	14	2.8	8.3	1.4	1	0.6	(9.70)	(6.85)	8.10	25	-	18.2	33 000	51 600
*35BER10X	35	62	14	2.8	8.3	1.4	1	0.6	(9.70)	(6.85)	8.10	25	-	18.2	39 200	61 900
*35BNR20SV1V	35	62	17	–	–	–	1	0.6	10.1	7.10	10.2	18	10.6	16.3	28 900	-
*35BNR20HV1V	35	62	17	–	–	–	1	0.6	(10.1)	(7.10)	6.70	18	10.6	16.3	37 200	-
*35BNR20XV1V	35	62	17	–	–	–	1	0.6	(10.1)	(7.10)	6.70	18	10.6	16.3	43 300	-
*35BER20SV1V	35	62	17	–	–	–	1	0.6	9.70	6.85	12.0	25	-	19.7	24 800	-
*35BER20HV1V	35	62	17	–	–	–	1	0.6	(9.70)	(6.85)	8.10	25	-	19.7	33 000	-
*35BER20XV1V	35	62	17	–	–	–	1	0.6	(9.70)	(6.85)	8.10	25	-	19.7	39 200	-
7207C	35	72	17	–	–	–	1.1	0.6	32.0	19.9	14.4	15	13.9	15.7	21 500	32 800
7207CSN24	35	72	17	–	–	–	1.1	0.6	(32.0)	(19.9)	17.1	15	13.9	15.7	28 100	42 700
7207A5	35	72	17	–	–	–	1.1	0.6	30.5	19.1	16.6	25	-	21.0	18 700	28 100
7207A5SN24	35	72	17	–	–	–	1.1	0.6	(30.5)	(19.1)	19.7	25	-	21.0	24 300	36 500
7207A	35	72	17	–	–	–	1.1	0.6	29.6	18.5	12.7	30	-	23.9	14 100	18 700

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(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트 Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
40	50	52.5	0.6	0.3	33	67	149	297	29	39	55	77	2	− 2	− 9	− 18	0.074
40	50	52.5	0.6	0.3	32	69	162	332	32	43	63	89	2	− 2	− 9	− 18	0.065
40	50	52.5	0.6	0.3	49	110	248	508	70	93	127	169	0	− 3	− 8	− 15	0.075
40	50	52.5	0.6	0.3	49	119	278	580	79	107	147	198	0	− 3	− 8	− 15	0.066
40	50	52.5	0.6	0.3	49	154	324	–	41	64	87	–	0	− 8	− 17	–	0.080
40	50	52.5	0.6	0.3	49	168	364	–	46	73	101	–	0	− 8	− 17	–	0.074
40	50	52.5	0.6	0.3	49	235	472	–	67	117	153	–	0	− 8	− 15	–	0.080
40	50	52.5	0.6	0.3	49	263	538	–	75	135	178	–	0	− 8	− 15	–	0.074
40	50	52.5	0.6	0.3	49	142	288	–	37	56	74	–	0	− 8	− 17	–	0.072
40	50	52.5	0.6	0.3	49	155	323	–	42	64	86	–	0	− 8	− 17	–	0.063
40	50	52.5	0.6	0.3	49	155	323	–	42	64	86	–	0	− 8	− 17	–	0.063
40	50	52.5	0.6	0.3	49	214	419	–	61	102	132	–	0	− 8	− 15	–	0.072
40	50	52.5	0.6	0.3	49	238	477	–	68	118	154	–	0	− 8	− 15	–	0.063
40	50	52.5	0.6	0.3	49	238	477	–	68	118	154	–	0	− 8	− 15	–	0.063
41	56	57	1	0.5	58	121	251	493	36	49	67	94	− 1	− 7	− 16	− 28	0.151
41	56	57	1	0.5	59	130	279	558	40	55	78	109	− 1	− 7	− 16	− 28	0.133
41	56	57	1	0.5	68	161	387	779	78	107	150	199	− 1	− 5	− 12	− 21	0.151
41	56	57	1	0.5	70	177	439	897	89	124	174	233	− 1	− 5	− 12	− 21	0.133
41	56	57	1	0.5	49	203	421	693	93	151	197	237	0	− 5	− 10	− 15	0.153
41	56	57	1	0.5	49	110	222	–	41	55	73	–	0	− 5	− 12	–	0.164
41	56	57	1	0.5	49	117	247	–	46	63	85	–	0	− 5	− 12	–	0.154
41	56	57	1	0.5	49	117	247	–	46	63	85	–	0	− 5	− 12	–	0.154
41	56	57	1	0.5	49	237	474	–	68	118	153	–	0	− 8	− 15	–	0.164
41	56	57	1	0.5	49	265	541	–	76	136	178	–	0	− 8	− 15	–	0.154
41	56	57	1	0.5	49	265	541	–	76	136	178	–	0	− 8	− 15	–	0.154
41	56	57	1	0.5	49	110	222	–	41	55	73	–	0	− 5	− 12	–	0.197
41	56	57	1	0.5	49	117	247	–	46	63	85	–	0	− 5	− 12	–	0.187
41	56	57	1	0.5	49	117	247	–	46	63	85	–	0	− 5	− 12	–	0.187
41	56	57	1	0.5	49	237	474	–	68	118	153	–	0	− 8	− 15	–	0.197
41	56	57	1	0.5	49	265	541	–	76	136	178	–	0	− 8	− 15	–	0.187
41	56	57	1	0.5	49	265	541	–	76	136	178	–	0	− 8	− 15	–	0.187
42	65	67	1	0.6	75	151	385	794	37	50	75	107	− 3	− 10	− 25	− 43	0.280
42	65	67	1	0.6	78	164	434	911	42	57	87	125	− 3	− 10	− 25	− 43	0.240
42	65	67	1	0.6	131	238	596	1 178	95	118	167	220	− 4	− 8	− 18	− 30	0.277
42	65	67	1	0.6	142	266	683	1 366	109	137	196	259	− 4	− 8	− 18	− 30	0.237
42	65	67	1	0.6	49	401	654	1 288	90	184	221	286	0	− 10	− 15	− 25	0.284

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

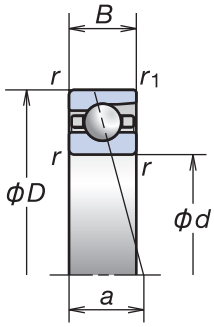
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

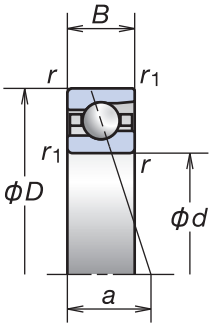
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

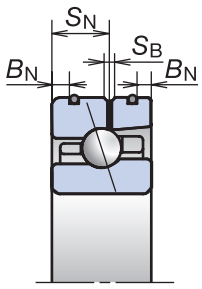
내경40mm



79,70,72계열
BNR10,20계열
BER10,20계열



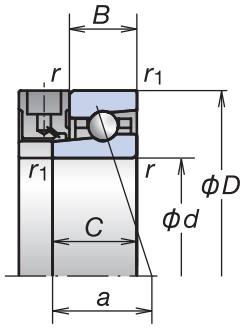
BNR19,29계열
BER19,29계열



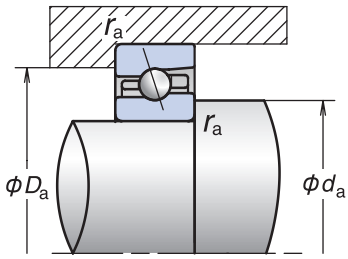
로바스트 쇼트

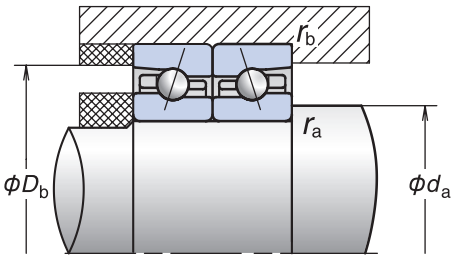
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7908C	40	62	12	–	–	–	–	0.6	0.3	15.9	11.7	8.40	15	15.7	12.8	22 600	34 400
*7908CSN24	40	62	12	–	–	–	–	0.6	0.3	(15.9)	(11.7)	9.97	15	15.7	12.8	29 500	44 800
*7908A5	40	62	12	–	–	–	–	0.6	0.3	15.0	11.2	8.90	25	–	17.9	19 700	29 500
*7908A5SN24	40	62	12	–	–	–	–	0.6	0.3	(15.0)	(11.2)	10.5	25	–	17.9	25 500	38 300
*40BNR19S	40	62	12	–	–	–	–	0.6	0.3	11.5	7.65	10.8	18	10.4	14.3	27 500	39 300
*40BNR19H	40	62	12	–	2.2	7.0	1.4	0.6	0.3	(11.5)	(7.65)	7.10	18	10.4	14.3	35 300	55 000
*40BNR19X	40	62	12	17	2.2	7.0	1.4	0.6	0.3	(11.5)	(7.65)	7.10	18	10.4	14.3	41 200	64 800
*40BER19S	40	62	12	–	–	–	–	0.6	0.3	11.0	7.35	12.8	25	–	17.9	23 600	33 400
*40BER19H	40	62	12	–	2.2	7.0	1.4	0.6	0.3	(11.0)	(7.35)	8.65	25	–	17.9	31 400	49 100
*40BER19X	40	62	12	17	2.2	7.0	1.4	0.6	0.3	(11.0)	(7.35)	8.65	25	–	17.9	37 300	58 900
*40BNR29SV1V	40	62	14	–	–	–	–	0.6	0.3	11.5	7.65	10.8	18	10.4	15.3	27 500	–
*40BNR29HV1V	40	62	14	–	–	–	–	0.6	0.3	(11.5)	(7.65)	7.10	18	10.4	15.3	35 300	–
*40BNR29XV1V	40	62	14	–	–	–	–	0.6	0.3	(11.5)	(7.65)	7.10	18	10.4	15.3	41 200	–
*40BER29SV1V	40	62	14	–	–	–	–	0.6	0.3	11.0	7.35	12.8	25	–	18.9	23 600	–
*40BER29HV1V	40	62	14	–	–	–	–	0.6	0.3	(11.0)	(7.35)	8.65	25	–	18.9	31 400	–
*40BER29XV1V	40	62	14	–	–	–	–	0.6	0.3	(11.0)	(7.35)	8.65	25	–	18.9	37 300	–
*7008C	40	68	15	–	–	–	–	1	0.6	21.6	15.9	10.6	15	15.4	14.7	21 300	32 500
*7008CSN24	40	68	15	–	–	–	–	1	0.6	(21.6)	(15.9)	12.5	15	15.4	14.7	27 800	42 300
*7008A5	40	68	15	–	–	–	–	1	0.6	20.5	15.1	12.0	25	–	20.1	18 600	27 800
*7008A5SN24	40	68	15	–	–	–	–	1	0.6	(20.5)	(15.1)	14.2	25	–	20.1	24 100	36 200
*7008A	40	68	15	–	–	–	–	1	0.6	19.7	14.6	9.15	30	–	23.1	13 900	18 600
*40BNR10S	40	68	15	–	–	–	–	1	0.6	10.6	7.95	11.5	18	10.7	16.2	26 000	37 100
*40BNR10H	40	68	15	–	2.8	8.8	1.4	1	0.6	(10.6)	(7.95)	7.50	18	10.7	16.2	33 400	51 900
*40BNR10X	40	68	15	20	2.8	8.8	1.4	1	0.6	(10.6)	(7.95)	7.50	18	10.7	16.2	38 900	61 200
*40BER10S	40	68	15	–	–	–	–	1	0.6	10.1	7.65	13.5	25	–	19.9	22 300	31 500
*40BER10H	40	68	15	–	2.8	8.8	1.4	1	0.6	(10.1)	(7.65)	9.10	25	–	19.9	29 700	46 300
*40BER10X	40	68	15	20	2.8	8.8	1.4	1	0.6	(10.1)	(7.65)	9.10	25	–	19.9	35 200	55 600
*40BNR20SV1V	40	68	18	–	–	–	–	1	0.6	10.6	7.95	11.5	18	10.7	17.7	26 000	–
*40BNR20HV1V	40	68	18	–	–	–	–	1	0.6	(10.6)	(7.95)	7.50	18	10.7	17.7	33 400	–
*40BNR20XV1V	40	68	18	–	–	–	–	1	0.6	(10.6)	(7.95)	7.50	18	10.7	17.7	38 900	–
*40BER20SV1V	40	68	18	–	–	–	–	1	0.6	10.1	7.65	13.5	25	–	21.4	22 300	–
*40BER20HV1V	40	68	18	–	–	–	–	1	0.6	(10.1)	(7.65)	9.10	25	–	21.4	29 700	–
*40BER20XV1V	40	68	18	–	–	–	–	1	0.6	(10.1)	(7.65)	9.10	25	–	21.4	35 200	–
7208C	40	80	18	–	–	–	–	1.1	0.6	38.0	25.2	17.6	15	14.1	17.0	19 200	29 200
7208CSN24	40	80	18	–	–	–	–	1.1	0.6	(38.0)	(25.2)	20.9	15	14.1	17.0	25 000	38 000
7208A5	40	80	18	–	–	–	–	1.1	0.6	36.5	24.1	20.6	25	–	23.0	16 700	25 000
7208A5SN24	40	80	18	–	–	–	–	1.1	0.6	(36.5)	(24.1)	24.4	25	–	23.0	21 700	32 500
7208A	40	80	18	–	–	–	–	1.1	0.6	35.5	23.3	15.8	30	–	26.3	12 500	16 700

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
45	57	59.5	0.6	0.3	41	78	196	384	32	42	63	88	1	− 3	− 12	− 22	0.109
45	57	59.5	0.6	0.3	40	81	215	432	35	47	72	101	1	− 3	− 12	− 22	0.096
45	57	59.5	0.6	0.3	68	113	291	572	81	97	139	182	− 1	− 3	− 9	− 16	0.110
45	57	59.5	0.6	0.3	71	121	327	655	92	111	161	213	− 1	− 3	− 9	− 16	0.097
45	57	59.5	0.6	0.3	49	145	277	–	38	57	74	–	0	− 8	− 16	–	0.105
45	57	59.5	0.6	0.3	49	158	310	–	43	66	86	–	0	− 8	− 16	–	0.092
45	57	59.5	0.6	0.3	49	158	310	–	43	66	86	–	0	− 8	− 16	–	0.092
45	57	59.5	0.6	0.3	49	221	434	–	63	106	137	–	0	− 8	− 15	–	0.105
45	57	59.5	0.6	0.3	49	246	494	–	71	123	160	–	0	− 8	− 15	–	0.092
45	57	59.5	0.6	0.3	49	246	494	–	71	123	160	–	0	− 8	− 15	–	0.092
45	57	59.5	0.6	0.3	49	145	277	–	38	57	74	–	0	− 8	− 16	–	0.120
45	57	59.5	0.6	0.3	49	158	310	–	43	66	86	–	0	− 8	− 16	–	0.107
45	57	59.5	0.6	0.3	49	158	310	–	43	66	86	–	0	− 8	− 16	–	0.107
45	57	59.5	0.6	0.3	49	221	434	–	63	106	137	–	0	− 8	− 15	–	0.120
45	57	59.5	0.6	0.3	49	246	494	–	71	123	160	–	0	− 8	− 15	–	0.107
45	57	59.5	0.6	0.3	49	246	494	–	71	123	160	–	0	− 8	− 15	–	0.107
46	62	63	1	0.5	58	114	291	594	39	51	77	110	− 1	− 6	− 17	− 30	0.189
46	62	63	1	0.5	59	123	325	676	43	58	89	128	− 1	− 6	− 17	− 30	0.168
46	62	63	1	0.5	92	203	424	864	95	127	167	223	− 2	− 6	− 12	− 21	0.188
46	62	63	1	0.5	98	225	483	998	108	147	195	262	− 2	− 6	− 12	− 21	0.167
46	62	63	1	0.5	49	219	463	768	101	168	221	267	0	− 5	− 10	− 15	0.191
46	62	63	1	0.5	49	114	216	–	44	60	77	–	0	− 5	− 11	–	0.204
46	62	63	1	0.5	49	122	240	–	49	68	89	–	0	− 5	− 11	–	0.193
46	62	63	1	0.5	49	122	240	–	49	68	89	–	0	− 5	− 11	–	0.193
46	62	63	1	0.5	49	252	510	–	72	128	167	–	0	− 8	− 15	–	0.204
46	62	63	1	0.5	49	282	583	–	81	148	195	–	0	− 8	− 15	–	0.193
46	62	63	1	0.5	49	282	583	–	81	148	195	–	0	− 8	− 15	–	0.193
46	62	63	1	0.5	49	114	216	–	44	60	77	–	0	− 5	− 11	–	0.242
46	62	63	1	0.5	49	122	240	–	49	68	89	–	0	− 5	− 11	–	0.231
46	62	63	1	0.5	49	122	240	–	49	68	89	–	0	− 5	− 11	–	0.231
46	62	63	1	0.5	49	252	510	–	72	128	167	–	0	− 8	− 15	–	0.242
46	62	63	1	0.5	49	282	583	–	81	148	195	–	0	− 8	− 15	–	0.231
46	62	63	1	0.5	49	282	583	–	81	148	195	–	0	− 8	− 15	–	0.231
47	73	75	1	0.6	98	202	501	985	44	60	90	125	− 5	− 13	− 29	− 47	0.366
47	73	75	1	0.6	104	223	570	1 133	50	69	104	147	− 5	− 13	− 29	− 47	0.313
47	73	75	1	0.6	138	290	750	1 490	105	137	196	259	− 4	− 9	− 20	− 33	0.362
47	73	75	1	0.6	151	326	864	1 734	121	159	230	305	− 4	− 9	− 20	− 33	0.309
47	73	75	1	0.6	49	438	721	1 428	97	205	246	318	0	− 10	− 15	− 25	0.370

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

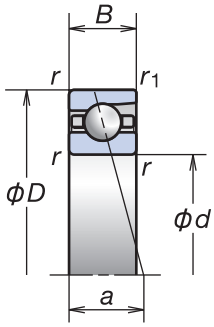
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

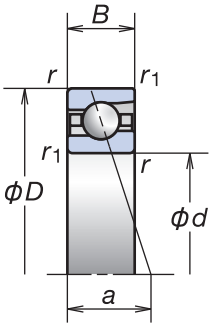
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

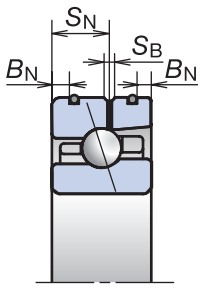
내경45mm



79,70,72계열
BNR10,20계열
BER10,20계열



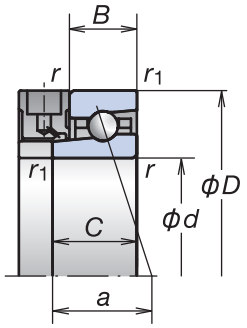
BNR19,29계열
BER19,29계열



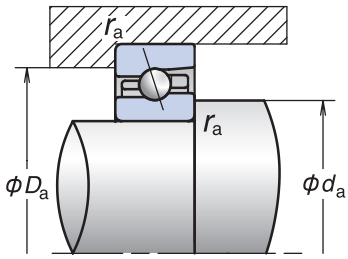
로바스트 쇼트

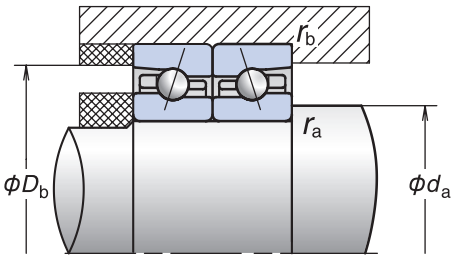
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7909C	45	68	12	–	–	–	–	0.6	0.3	16.8	13.4	8.55	15	16.0	13.6	20 400	31 000
*7909CSN24	45	68	12	–	–	–	–	0.6	0.3	(16.8)	(13.4)	10.1	15	16.0	13.6	26 600	40 400
*7909A5	45	68	12	–	–	–	–	0.6	0.3	15.9	12.7	9.95	25	–	19.2	17 700	26 600
*7909A5SN24	45	68	12	–	–	–	–	0.6	0.3	(15.9)	(12.7)	11.8	25	–	19.2	23 100	34 600
*45BNR19S	45	68	12	–	–	–	–	0.6	0.3	12.1	8.70	12.4	18	10.6	15.2	24 800	35 400
*45BNR19H	45	68	12	–	2.2	7.0	1.4	0.6	0.3	(12.1)	(8.70)	8.10	18	10.6	15.2	31 900	49 600
*45BNR19X	45	68	12	17	2.2	7.0	1.4	0.6	0.3	(12.1)	(8.70)	8.10	18	10.6	15.2	37 200	58 500
*45BER19S	45	68	12	–	–	–	–	0.6	0.3	11.6	8.35	14.6	25	–	19.2	21 300	30 100
*45BER19H	45	68	12	–	2.2	7.0	1.4	0.6	0.3	(11.6)	(8.35)	9.85	25	–	19.2	28 400	44 300
*45BER19X	45	68	12	17	2.2	7.0	1.4	0.6	0.3	(11.6)	(8.35)	9.85	25	–	19.2	33 700	53 100
*45BNR29SV1V	45	68	14	–	–	–	–	0.6	0.3	12.1	8.70	12.4	18	10.6	16.2	24 800	–
*45BNR29HV1V	45	68	14	–	–	–	–	0.6	0.3	(12.1)	(8.70)	8.10	18	10.6	16.2	31 900	–
*45BNR29XV1V	45	68	14	–	–	–	–	0.6	0.3	(12.1)	(8.70)	8.10	18	10.6	16.2	37 200	–
*45BER29SV1V	45	68	14	–	–	–	–	0.6	0.3	11.6	8.35	14.6	25	–	20.2	21 300	–
*45BER29HV1V	45	68	14	–	–	–	–	0.6	0.3	(11.6)	(8.35)	9.85	25	–	20.2	28 400	–
*45BER29XV1V	45	68	14	–	–	–	–	0.6	0.3	(11.6)	(8.35)	9.85	25	–	20.2	33 700	–
*7009C	45	75	16	–	–	–	–	1	0.6	25.6	19.3	12.4	15	15.4	16.0	19 200	29 200
*7009CSN24	45	75	16	–	–	–	–	1	0.6	(25.6)	(19.3)	14.7	15	15.4	16.0	25 000	38 000
*7009A5	45	75	16	–	–	–	–	1	0.6	24.3	18.3	14.5	25	–	22.0	16 700	25 000
*7009A5SN24	45	75	16	–	–	–	–	1	0.6	(24.3)	(18.3)	17.2	25	–	22.0	21 700	32 500
*7009A	45	75	16	–	–	–	–	1	0.6	23.4	17.7	11.1	30	–	25.3	12 500	16 700
*45BNR10E	45	75	16	–	–	–	–	1	0.6	11.7	9.00	12.7	18	10.6	17.6	25 000	35 700
*45BNR10H	45	75	16	–	3.4	9.3	1.4	1	0.6	(11.7)	(9.00)	8.35	18	10.6	17.6	30 000	46 700
*45BNR10X	45	75	16	21	3.4	9.3	1.4	1	0.6	(11.7)	(9.00)	8.35	18	10.6	17.6	35 000	55 000
*45BER10E	45	75	16	–	–	–	–	1	0.6	11.2	8.60	15.0	25	–	21.8	21 500	30 400
*45BER10H	45	75	16	–	3.4	9.3	1.4	1	0.6	(11.2)	(8.60)	10.1	25	–	21.8	26 700	41 700
*45BER10X	45	75	16	21	3.4	9.3	1.4	1	0.6	(11.2)	(8.60)	10.1	25	–	21.8	31 700	50 000
*45BNR20EV1V	45	75	19	–	–	–	–	1	0.6	11.7	9.00	12.7	18	10.6	19.1	25 000	–
*45BNR20HV1V	45	75	19	–	–	–	–	1	0.6	(11.7)	(9.00)	8.35	18	10.6	19.1	30 000	–
*45BNR20XV1V	45	75	19	–	–	–	–	1	0.6	(11.7)	(9.00)	8.35	18	10.6	19.1	35 000	–
*45BER20EV1V	45	75	19	–	–	–	–	1	0.6	11.2	8.60	15.0	25	–	23.3	21 500	–
*45BER20HV1V	45	75	19	–	–	–	–	1	0.6	(11.2)	(8.60)	10.1	25	–	23.3	26 700	–
*45BER20XV1V	45	75	19	–	–	–	–	1	0.6	(11.2)	(8.60)	10.1	25	–	23.3	31 700	–
7209C	45	85	19	–	–	–	–	1.1	0.6	43.0	28.8	19.6	15	14.2	18.2	17 700	27 000
7209CSN24	45	85	19	–	–	–	–	1.1	0.6	(43.0)	(28.8)	23.3	15	14.2	18.2	23 100	35 100
7209A5	45	85	19	–	–	–	–	1.1	0.6	41.0	27.6	23.3	25	–	24.7	15 400	23 100
7209A5SN24	45	85	19	–	–	–	–	1.1	0.6	(41.0)	(27.6)	27.7	25	–	24.7	20 000	30 000
7209A	45	85	19	–	–	–	–	1.1	0.6	39.5	26.7	18.0	30	–	28.3	11 600	15 400

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트 Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
50	63	65.5	0.6	0.3	49	104	192	391	37	51	66	93	0	− 5	− 11	− 21	0.129
50	63	65.5	0.6	0.3	49	111	211	440	41	57	76	108	0	− 5	− 11	− 21	0.115
50	63	65.5	0.6	0.3	70	146	353	676	88	115	160	208	− 1	− 4	− 10	− 17	0.130
50	63	65.5	0.6	0.3	72	159	399	776	99	132	187	243	− 1	− 4	− 10	− 17	0.116
50	63	65.5	0.6	0.3	49	153	298	–	41	63	82	–	0	− 8	− 16	–	0.125
50	63	65.5	0.6	0.3	49	168	335	–	46	72	95	–	0	− 8	− 16	–	0.111
50	63	65.5	0.6	0.3	49	168	335	–	46	72	95	–	0	− 8	− 16	–	0.111
50	63	65.5	0.6	0.3	49	237	473	–	68	117	151	–	0	− 8	− 15	–	0.125
50	63	65.5	0.6	0.3	49	265	539	–	76	136	177	–	0	− 8	− 15	–	0.111
50	63	65.5	0.6	0.3	49	265	539	–	76	136	177	–	0	− 8	− 15	–	0.111
50	63	65.5	0.6	0.3	49	153	298	–	41	63	82	–	0	− 8	− 16	–	0.143
50	63	65.5	0.6	0.3	49	168	335	–	46	72	95	–	0	− 8	− 16	–	0.128
50	63	65.5	0.6	0.3	49	168	335	–	46	72	95	–	0	− 8	− 16	–	0.128
50	63	65.5	0.6	0.3	49	237	473	–	68	117	151	–	0	− 8	− 15	–	0.143
50	63	65.5	0.6	0.3	49	265	539	–	76	136	177	–	0	− 8	− 15	–	0.128
50	63	65.5	0.6	0.3	49	265	539	–	76	136	177	–	0	− 8	− 15	–	0.128
51	69	70	1	0.5	80	144	338	695	45	57	84	120	− 3	− 8	− 19	− 33	0.238
51	69	70	1	0.5	84	156	380	794	51	66	97	140	− 3	− 8	− 19	− 33	0.211
51	69	70	1	0.5	94	210	485	958	99	132	181	238	− 2	− 6	− 13	− 22	0.250
51	69	70	1	0.5	99	233	553	1 107	113	153	212	280	− 2	− 6	− 13	− 22	0.223
51	69	70	1	0.5	49	227	482	1 178	105	176	231	324	0	− 5	− 10	− 20	0.241
51	69	70	1	0.5	49	114	218	–	44	60	77	–	0	− 5	− 11	–	0.259
51	69	70	1	0.5	49	123	242	–	50	69	89	–	0	− 5	− 11	–	0.246
51	69	70	1	0.5	49	123	242	–	50	69	89	–	0	− 5	− 11	–	0.246
51	69	70	1	0.5	49	255	516	–	73	129	168	–	0	− 8	− 15	–	0.259
51	69	70	1	0.5	49	285	590	–	82	150	197	–	0	− 8	− 15	–	0.246
51	69	70	1	0.5	49	285	590	–	82	150	197	–	0	− 8	− 15	–	0.246
51	69	70	1	0.5	49	114	218	–	44	60	77	–	0	− 5	− 11	–	0.305
51	69	70	1	0.5	49	123	242	–	50	69	89	–	0	− 5	− 11	–	0.291
51	69	70	1	0.5	49	123	242	–	50	69	89	–	0	− 5	− 11	–	0.291
51	69	70	1	0.5	49	255	516	–	73	129	168	–	0	− 8	− 15	–	0.305
51	69	70	1	0.5	49	285	590	–	82	150	197	–	0	− 8	− 15	–	0.291
51	69	70	1	0.5	49	285	590	–	82	150	197	–	0	− 8	− 15	–	0.291
52	78	80	1	0.6	123	254	534	1 067	49	67	94	131	− 7	− 16	− 30	− 49	0.406
52	78	80	1	0.6	132	283	608	1 230	56	77	109	154	− 7	− 16	− 30	− 49	0.341
52	78	80	1	0.6	169	333	875	1 738	115	147	212	281	− 5	− 10	− 22	− 36	0.402
52	78	80	1	0.6	186	377	1 011	2 026	133	171	249	330	− 5	− 10	− 22	− 36	0.337
52	78	80	1	0.6	49	450	1 084	1 905	100	212	292	363	0	− 10	− 20	− 30	0.410

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

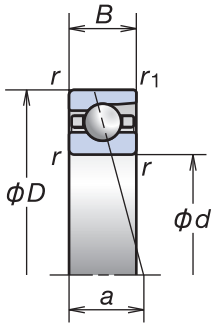
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

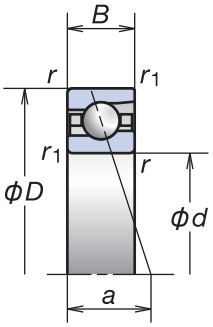
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

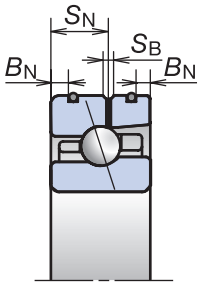
내경50mm



79,70,72계열
BNR10,20계열
BER10,20계열



BNR19,29계열
BER19,29계열



로바스트 쇼트

호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7910C	50	72	12	–	–	–	–	0.6	0.3	17.7	15.0	9.45	15	16.2	14.2	18 900	28 700
*7910CSN24	50	72	12	–	–	–	–	0.6	0.3	(17.7)	(15.0)	11.2	15	16.2	14.2	24 600	37 400
*7910A5	50	72	12	–	–	–	–	0.6	0.3	16.7	14.2	11.0	25	–	20.2	16 400	24 600
*7910A5SN24	50	72	12	–	–	–	–	0.6	0.3	(16.7)	(14.2)	13.1	25	–	20.2	21 400	32 000
*50BNR19S	50	72	12	–	–	–	–	0.6	0.3	12.8	9.75	13.9	18	10.7	15.9	23 000	32 800
*50BNR19H	50	72	12	–	2.2	7.0	1.4	0.6	0.3	(12.8)	(9.75)	9.10	18	10.7	15.9	29 600	46 000
*50BNR19X	50	72	12	17	2.2	7.0	1.4	0.6	0.3	(12.8)	(9.75)	9.10	18	10.7	15.9	34 500	54 100
*50BER19S	50	72	12	–	–	–	–	0.6	0.3	12.2	9.35	16.3	25	–	20.2	19 700	27 900
*50BER19H	50	72	12	–	2.2	7.0	1.4	0.6	0.3	(12.2)	(9.35)	11.0	25	–	20.2	26 300	41 000
*50BER19X	50	72	12	17	2.2	7.0	1.4	0.6	0.3	(12.2)	(9.35)	11.0	25	–	20.2	31 200	49 200
*50BNR29SV1V	50	72	14	–	–	–	–	0.6	0.3	12.8	9.75	13.9	18	10.7	16.9	23 000	–
*50BNR29HV1V	50	72	14	–	–	–	–	0.6	0.3	(12.8)	(9.75)	9.10	18	10.7	16.9	29 600	–
*50BNR29XV1V	50	72	14	–	–	–	–	0.6	0.3	(12.8)	(9.75)	9.10	18	10.7	16.9	34 500	–
*50BER29SV1V	50	72	14	–	–	–	–	0.6	0.3	12.2	9.35	16.3	25	–	21.2	19 700	–
*50BER29HV1V	50	72	14	–	–	–	–	0.6	0.3	(12.2)	(9.35)	11.0	25	–	21.2	26 300	–
*50BER29XV1V	50	72	14	–	–	–	–	0.6	0.3	(12.2)	(9.35)	11.0	25	–	21.2	31 200	–
*7010C	50	80	16	–	–	–	–	1	0.6	27.3	21.9	13.9	15	15.7	16.7	17 700	27 000
*7010CSN24	50	80	16	–	–	–	–	1	0.6	(27.3)	(21.9)	16.5	15	15.7	16.7	23 100	35 100
*7010A5	50	80	16	–	–	–	–	1	0.6	25.8	20.8	16.2	25	–	23.2	15 400	23 100
*7010A5SN24	50	80	16	–	–	–	–	1	0.6	(25.8)	(20.8)	19.3	25	–	23.2	20 000	30 000
*7010A	50	80	16	–	–	–	–	1	0.6	24.8	20.1	12.5	30	–	26.8	11 600	15 400
*50BNR10E	50	80	16	–	–	–	–	1	0.6	12.2	9.90	14.0	18	10.8	18.4	23 100	33 000
*50BNR10H	50	80	16	–	3.4	9.3	1.4	1	0.6	(12.2)	(9.90)	9.20	18	10.8	18.4	27 700	43 100
*50BNR10X	50	80	16	21	3.4	9.3	1.4	1	0.6	(12.2)	(9.90)	9.20	18	10.8	18.4	32 400	50 800
*50BER10E	50	80	16	–	–	–	–	1	0.6	11.6	9.50	16.5	25	–	23.0	19 900	28 000
*50BER10H	50	80	16	–	3.4	9.3	1.4	1	0.6	(11.6)	(9.50)	11.1	25	–	23.0	24 700	38 500
*50BER10X	50	80	16	21	3.4	9.3	1.4	1	0.6	(11.6)	(9.50)	11.1	25	–	23.0	29 300	46 200
*50BNR20EV1V	50	80	19	–	–	–	–	1	0.6	12.2	9.90	14.0	18	10.8	19.9	23 100	–
*50BNR20HV1V	50	80	19	–	–	–	–	1	0.6	(12.2)	(9.90)	9.20	18	10.8	19.9	27 700	–
*50BNR20XV1V	50	80	19	–	–	–	–	1	0.6	(12.2)	(9.90)	9.20	18	10.8	19.9	32 400	–
*50BER20EV1V	50	80	19	–	–	–	–	1	0.6	11.6	9.50	16.5	25	–	24.5	19 900	–
*50BER20HV1V	50	80	19	–	–	–	–	1	0.6	(11.6)	(9.50)	11.1	25	–	24.5	24 700	–
*50BER20XV1V	50	80	19	–	–	–	–	1	0.6	(11.6)	(9.50)	11.1	25	–	24.5	29 300	–
7210C	50	90	20	–	–	–	–	1.1	0.6	45.0	31.5	21.1	15	14.5	19.4	16 500	25 000
7210CSN24	50	90	20	–	–	–	–	1.1	0.6	(45.0)	(31.5)	25.0	15	14.5	19.4	20 600	31 300
7210A5	50	90	20	–	–	–	–	1.1	0.6	43.0	30.5	25.2	25	–	26.3	14 300	21 500
7210A5SN24	50	90	20	–	–	–	–	1.1	0.6	(43.0)	(30.5)	29.9	25	–	26.3	17 900	26 900
7210A	50	90	20	–	–	–	–	1.1	0.6	41.5	29.3	19.4	30	–	30.2	10 800	14 300

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
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(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.